

MINJAE CHOI

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SUMMARY

Recent Data Science graduate with hands-on experience in data analysis, optimization, and applied AI projects. Experienced in transforming raw datasets into structured analyses, dashboards, and decision-support tools using Python, R, SQL, and Shiny. Strong analytical foundation with emphasis on problem formulation, metric design, and evidence-based decision making.

SKILLS

Programming & Analysis: Python (pandas, numpy, scikit-learn), R (dplyr, ggplot2, Shiny), SQL, Java, HTML

Data: Data cleaning, feature engineering, EDA, optimization logic

Tools: Excel (pivot tables, formulas), GitHub, relational databases, Linux (shell scripting), Docker, Unix pipelines

EXPERIENCE

University of Wisconsin–Madison CDIS

Madison, WI

AI Developer, SAIL (Summer AI Lab)

May 2025 – Aug 2025

- Implemented vector-database memory separating short-term context and long-term embeddings to improve response consistency
- Reduced repetitive NPC dialogue during internal testing by improving retrieval logic and prompt structure
- Collaborated via GitHub to debug gameplay logic and evaluate AI behavior across test scenarios
- Designed API pipelines to support AI-driven NPC dialogue and state management within a 2D RPG system

Memorial Union Catering Team

Madison, WI

Catering Staff

May 2024 – Aug 2024

- Supported high-volume campus events by coordinating setup, service, and breakdown under time constraints
- Communicated with staff and event coordinators to resolve on-site issues and maintain smooth operations

Korean Undergraduate Computer Science Union

Madison, WI

Active Member

Jan 2024 – May 2024

- Contributed to student-led website planning discussions, including feature ideas and basic structure
- Participated in technical talks and mentoring sessions focused on software engineering fundamentals

PROJECTS

College ROI Data Visualization & Analysis Project

Madison, WI

Project Leader

Aug 2025 – Dec 2025

- Analyzed data from 1,800+ U.S. colleges to evaluate relationships between tuition, debt, earnings, and ROI
- Cleaned and standardized multi-year College Scorecard data in R for cross-institution comparison
- Built interactive Shiny dashboards enabling users to explore ROI patterns by major and degree level
- Identified substantial ROI variation across fields of study, supporting more informed education investment decisions

FC25 Soccer Data Analysis & Optimization Project

Madison, WI

Project Member

Aug 2024 – Dec 2024

- Collected and cleaned player performance data using Python (pandas) and SQL
- Engineered team-level metrics to evaluate formation effectiveness based on player attributes
- Implemented optimization logic to compare formations and identify lineups maximizing defined performance metrics
- Used R for exploratory analysis and visualization to validate metric behavior

Healthcare Analytics SQL Project

- Built a normalized healthcare relational database and performed large-scale analytical queries using MySQL (CTEs, joins, window functions).
- Simulated real-world clinical and billing data to analyze provider productivity, patient utilization patterns, and revenue metrics.

EDUCATION

University of Wisconsin–Madison

Bachelor of Science, Data Science

May 2023 – Dec 2025

Relevant Courses: Data Science Modeling I&II, Database Design, Artificial Intelligence, Data Visualization

CERTIFICATION

Google Data Analytics Professional Certificate